Cambridge (CIE) A Level Maths: Probability & Statistics 1

Your notes

Probability Distributions

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Discrete Random Variables

What is a discrete random variable?

- A **random variable** is a variable whose value depends on the outcome of a **random event**
  - The value of the random variable is not known until the event is carried out (this is what is meant by 'random' in this case)
- **Random variables** are denoted using **upper case letters** (X , Y , etc)
- **Particular outcomes** of the event are denoted using **lower case letters** (x, y, etc)
- $\mathrm{P}(X = x)$ means "the probability of the random variable *X* taking the value x"
- A discrete random variable (often abbreviated to DRV) can only take **certain values** within a set
  - Discrete random variables **usually count** something
  - Discrete random variables usually can only take a finite number of values but it is possible that it can take an infinite number of values (see the examples below)
- **Examples** of discrete random variables include:
  - The number of times a coin lands on heads when flipped 20 times (this has a finite number of outcomes: 0,1,2,...,20)
  - The number of emails a manager receives within an hour (this has an infinite number of outcomes: 1,2,3,...)
  - The number of times a dice is rolled until it lands on a 6 (this has an infinite number of outcomes: 1,2,3,...)
  - The number on a bingo ball when one is drawn at random (this has a finite number of outcomes: 1,2,3...,90)

Probability Distributions (Discrete)

What is a probability distribution?

- A **discrete probability distribution** fully describes **all the values** that a discrete random variable can take along with their **associated probabilities**
  - This can be given in a **table**
  - Or it can be given as a **function** (called a probability mass function)
  - They can be represented by **vertical line graphs** (the possible values for *X* along the horizontal axis and the probability on the vertical axis)
- The **sum of the probabilities** of **all the values** of a discrete random variable is **1**
  - This is usually written $\Sigma \mathrm{P}(X = x) = 1$

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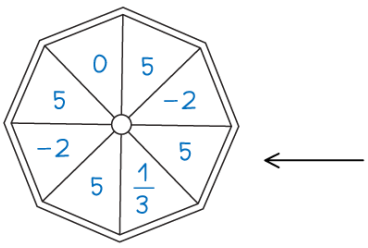
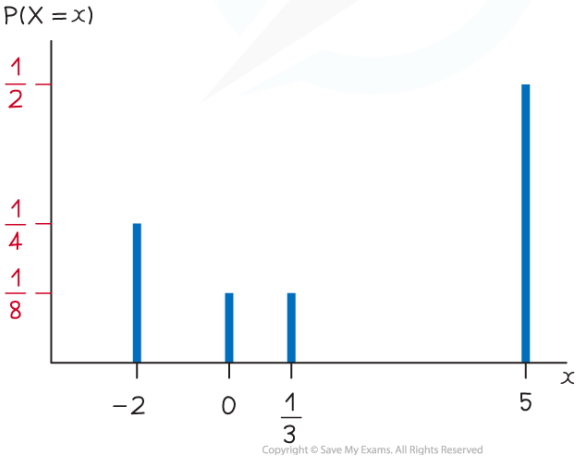


LET x BE THE NUMBER THAT THE SPINNER LANDS ON

x	-2	0	$\frac{1}{3}$	5
$P(X=x)$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{2}$

$$P(X=x) = \begin{cases} \frac{1}{8} & x = 0, \frac{1}{3} \\ \frac{1}{4} & x = -2 \\ \frac{1}{2} & x = 5 \\ 0 & \text{OTHERWISE} \end{cases}$$



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Cumulative Probabilities (Discrete)

How do I calculate probabilities using a discrete probability distribution?

- First **draw a table** to represent the probability distribution
  - If it is given as a function then find each probability
  - If any probabilities are unknown then use algebra to represent them
- **Form an equation** using $\sum P(X=x)=1$
  - Add together all the probabilities and make the sum equal to 1
- To find $P(X=k)$
  - If k is a possible value of the random variable X then $P(X=k)$ will be given in the table

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- If k is not a possible value then $P(X = k) = 0$
- To find $P(X \leq k)$
  - Identify all possible values, x_i, that X can take which satisfy $x_i \leq k$
  - Add together all their corresponding probabilities
  - $\mathrm{P}(X \leq k) = \sum_{x_i \leq k} \mathrm{P}(X = x_i)$
  - Some mathematicians use the notation F(x) to represent the cumulative distribution
    - $\mathrm{F}(x) = \mathrm{P}(X \leq x)$
- Using a similar method you can find $P(X < k)$, $P(X \geq k)$ and $P(X > k)$
- As all the probabilities add up to 1 you can form the following equivalent equations:
  - $\mathrm{P}(X < k) + \mathrm{P}(X = k) + \mathrm{P}(X > k) = 1$
  - $\mathrm{P}(X > k) = 1 - \mathrm{P}(X \leq k)$
  - $\mathrm{P}(X \geq k) = 1 - \mathrm{P}(X < k)$
- To calculate more complicated probabilities such as $\mathrm{P}(X^2 < 4)$
  - Identify which values of the random variable satisfy the inequality or event in the brackets
  - Add together the corresponding probabilities

How do I know which inequality to use?

- $\mathrm{P}(X \leq k)$ would be used for phrases such as:
  - At most k, no greater than k, etc
- $\mathrm{P}(X < k)$ would be used for phrases such as:
  - Fewer than k
- $\mathrm{P}(X \geq k)$ would be used for phrases such as:
  - At least k , no fewer than k, etc
- $\mathrm{P}(X > k)$ would be used for phrases such as:
  - Greater than k, etc

Worked Example

The probability distribution of the discrete random variable is given by the function

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$$P(X=x)=\begin{cases} kx^2 & x=-3,-1,2,4 \\ 0 & \text{otherwise.} \end{cases}$$

(a) Show that $k=\frac{1}{30}$.

(b) Calculate $P(X\le 3)$.

(c) Calculate $P(X^2<5)$

a) Construct a table

x	-3	-1	2	4
$P(X=x)$	$9k$	k	$4k$	$16k$

Substitute in the values of x
e.g. $P(X=-3)=k(-3)^2=9k$

The probabilities add up to 1

$9k+k+4k+16k=1$

$30k=1$

$k=\frac{1}{30}$

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b) Substitute k into the probabilities

x	-3	-1	2	4
$P(X=x)$	$\frac{3}{10}$	$\frac{1}{30}$	$\frac{2}{15}$	$\frac{8}{15}$

$X\leqslant 3$: $X=-3,-1,2$

$P(X\leqslant 3)=P(X=-3)+P(X=-1)+P(X=2)$

$=\frac{3}{10}+\frac{1}{30}+\frac{2}{15}$

$P(X\leqslant 3)=\frac{7}{15}$

Note: $1-\frac{8}{15}=\frac{7}{15}$

c) $X^2<5$ for $X=-1,2$

$P(X^2<5)=P(X=-1)+P(X=2)$

$=\frac{1}{30}+\frac{2}{15}$

$P(X^2<5)=\frac{1}{6}$

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Examiner Tips and Tricks

- Try to draw a table if there are a finite number of values that the discrete random variable can take
- When finding a probability, it will sometimes be quicker to subtract the probabilities of the unwanted values from 1 rather than adding together the probabilities of the wanted values
- Always make sure that the probabilities are between 0 and 1, and that they add up to 1!

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E(X) & Var(X) (Discrete)

What does E(X) mean and how do I calculate E(X)?

- **E(X)** means the **expected value** or the **mean** of a **random variable X**
- For a **discrete** random variable, it is calculated by:
  - **Multiplying each value** of X with its corresponding **probability**
  - **Adding** all these terms together

$$\Sigma x \mathrm{P}(X = x)$$

- Look out for **symmetrical** distributions (where the values of X are symmetrical and their probabilities are symmetrical) as the mean of these is the same as the median
  - For example if X can take the values 1, 5, 9 with probabilities 0.3, 0.4, 0.3 respectively then by symmetry the mean would be 5

How do I calculate E(X²)?

- **E(X²)** means the **expected value** or the **mean** of a **random variable** defined as **X²**
- For a **discrete** random variable, it is calculated by:
  - **Squaring** each value of X to get the values of X^2
  - **Multiplying each value** of X^2 with its corresponding **probability**
  - **Adding** all these terms together

$$\Sigma x^2 \mathrm{P}(X = x)$$

- In a similar way E(f(x)) can be calculated for a discrete random variable by:
  - **Applying the function f** to each value of to get the values of f(*X*)
  - **Multiplying each value** of f(*X*) with its corresponding **probability**
  - **Adding** all these terms together

$$\Sigma f(x)\, \mathrm{P}(X = x)$$

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x	a	b	c
$P(X=x)$	p	q	r

$$E(X) = ap + bq + cr$$

$$E(X^2) = a^2p + b^2q + c^2r$$

$$E\left(\frac{1}{X}\right) = \frac{p}{a} + \frac{q}{b} + \frac{r}{c}$$

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x	-2	3	5
$P(X=x)$	0.5	0.1	0.4

$$E(X) = (-2)(0.5) + (3)(0.1) + (5)(0.4)$$

$$E(X^2) = (-2)^2(0.5) + (3^2)(0.1) + (5^2)(0.4)$$

$$E\left(\frac{1}{X}\right) = \left(-\frac{1}{2}\right)(0.5) + \left(\frac{1}{3}\right)(0.1) + \left(\frac{1}{5}\right)(0.4)$$

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Is $E(X^2)$ equal to $(E(X))^2$?

- **Definitely not!**
  - They are only equal if X can take only one value with probability 1
    - if this was the case it would no longer be a random variable
- $E(X^2)$ is the **mean** of the values of $\mathbf{X^2}$
- $(E(X))^2$ is the **square** of the **mean** of the values of **X**
- To see the difference
  - Imagine a random variable X that can only take the values 1 and -1 with equal chance
  - The mean would be 0 so the square of the mean would also be 0
  - The square values would be 1 and 1 so the mean of the squares would also be 1
- In general $E(f(X))$ **does not equal** $f(E(X))$ where f is a function
  - So if you wanted to find something like $E\left(\frac{1}{x}\right)$ then you would have to use the definition and calculate:

$$\sum \frac{1}{x} \mathrm{P}(X = x)$$

Your notes

What does Var(X) mean and how do I calculate Var(X)?

- **Var(X)** means the **variance** of a **random variable X**
- For **any** random variable this can be calculated using the formula

$$E(X^2) - (E(X))^2$$

- This is the **mean of the squares of *X*** minus the **square of the mean of *X***
  - Compare this to the definition of the **variance of a set of data**
- Var(X) is always positive
- The **standard deviation** of a random variable X is the **square root** of **Var(X)**

Worked Example

The discrete random variable X has the probability distribution shown in the following table:

x	2	3	5	7
$\mathrm{P}(X = x)$	0.1	0.3	0.2	0.4

(a) Find the value of $\mathrm{E}(X)$.

(b) Find the value of $E(X^2)$.

(c) Find the value of $\mathrm{Var}(X)$.

a) Use the formula "E(x) = ∑xP(X = x)"

E(x) = (2)(0.1) + (3)(0.3) + (5)(0.2) + (7)(0.4)

E(x) = 4.9

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b) Use the formula "E(x²) = ∑x²P(X = x)"

E(x²) = (2²)(0.1) + (3²)(0.3) + (5²)(0.2) + (7²)(0.4)

E(x²) = 27.7

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c) Use the formula "$Var(x) = E(x^2) - (E(x))^2$"

$Var(x) = 27.7 - 4.9^2$

$Var(x) = 3.69$

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Examiner Tips and Tricks

- Check if your answer makes sense. The mean should fit within the range of the values of X.